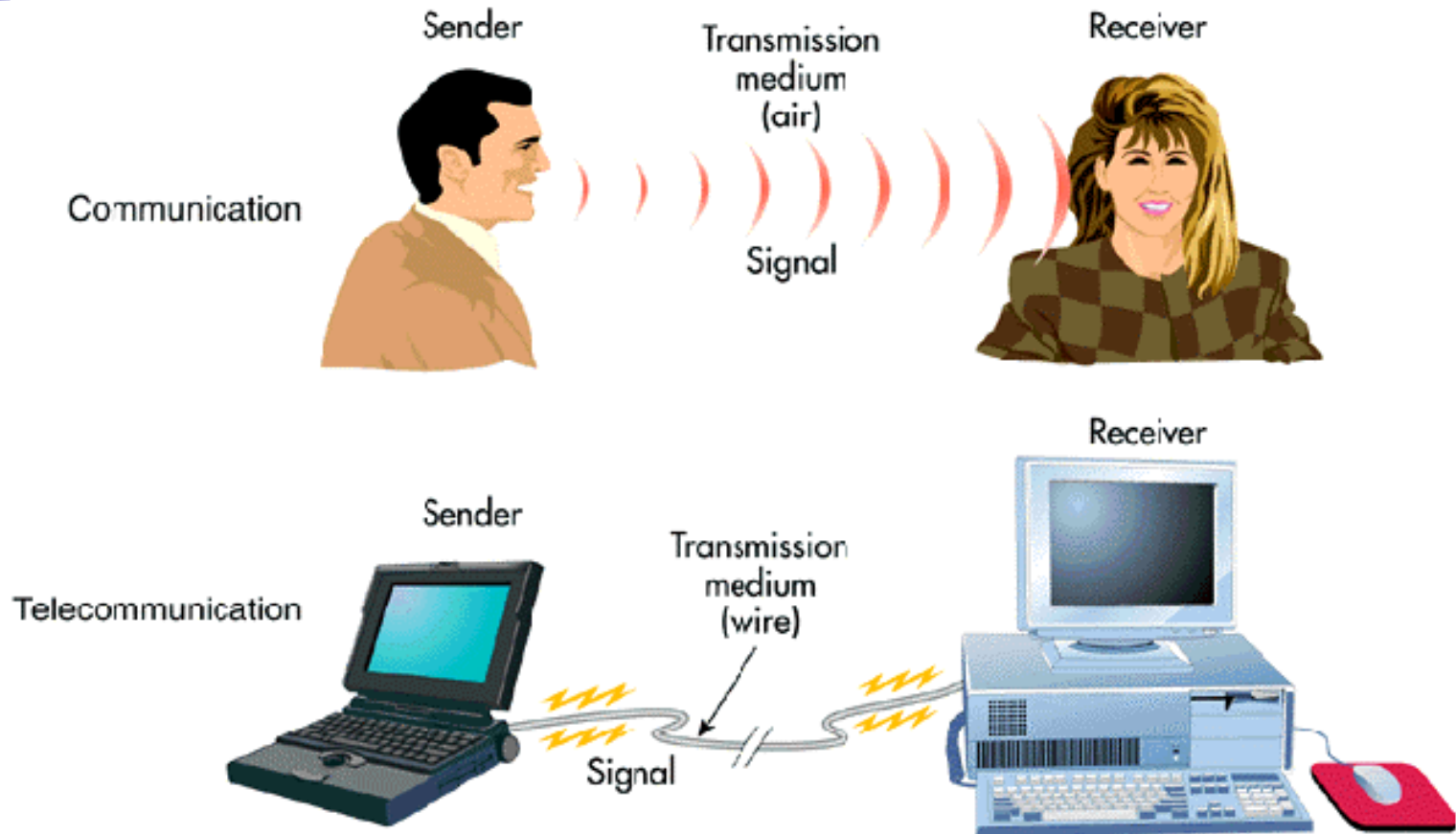


Lecture 1.1:

Evolution of Telecommunication

Introduction





Introduction

- **Calling Subscriber:** The one who initiates the call is referred to as the Calling Subscriber .
- **Called Subscriber:** The one for whom the call is destined is the Called Subscriber .

Evolution of Telecommunications

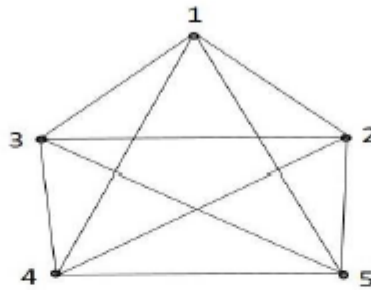
➤ **History and Growth** Telegraphy was introduced in 1837 in Great Britain and in 1845 in France.

➤ In March 1876, **Alexander Graham Bell** invented and demonstrated his telephone set and the possibility of long distance voice communication .



Evolution of Telecommunications

Alexander Graham Bell demonstrated the **point-to-point communication**

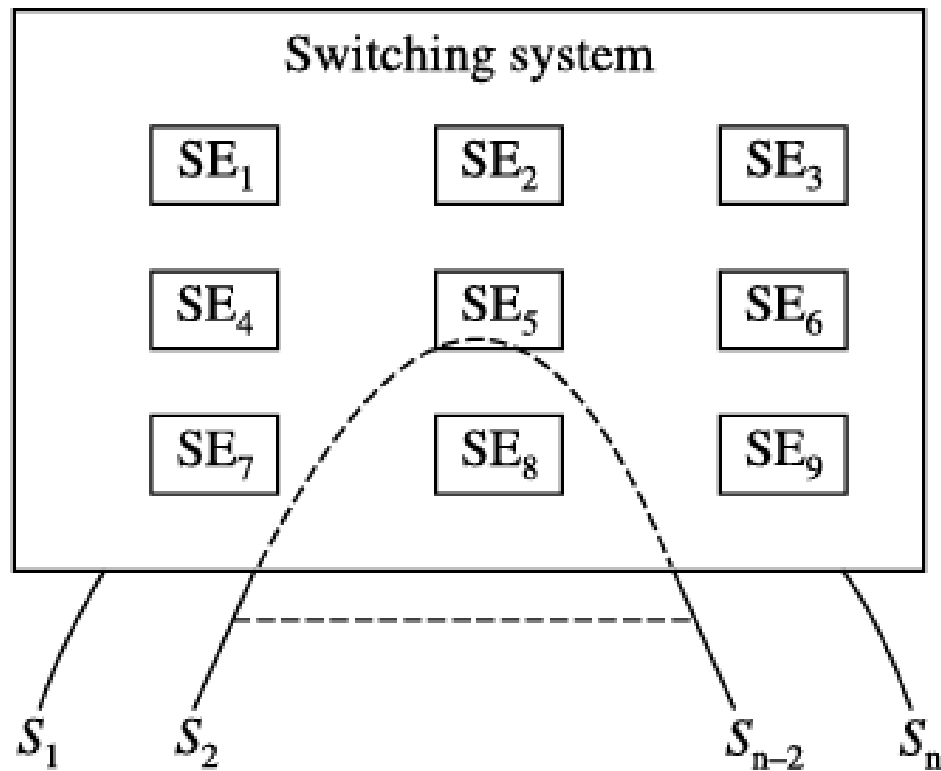
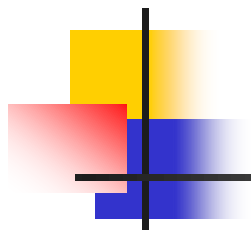


- The point-to-point connection for establishing communication requires the telephone sets to be linked using wires
- Point-to-point connection, for **n entities**, we need **$n(n-1)/2$ links**
- Networks with point-to-point links among all the entities are known as **Fully Connected Networks**



Switching Systems

- A switching system is variedly known as a switching office, a switching centre, a telephone exchange, a switching node or simply a switch.
- A switching system contains a number of switching elements.
- When a subscriber wants to communicate with another a connection is established between the two subscribers at the switching system by using one or more switching element.
- The functions performed by a switching system in establishing and releasing connections are known as call processing functions.



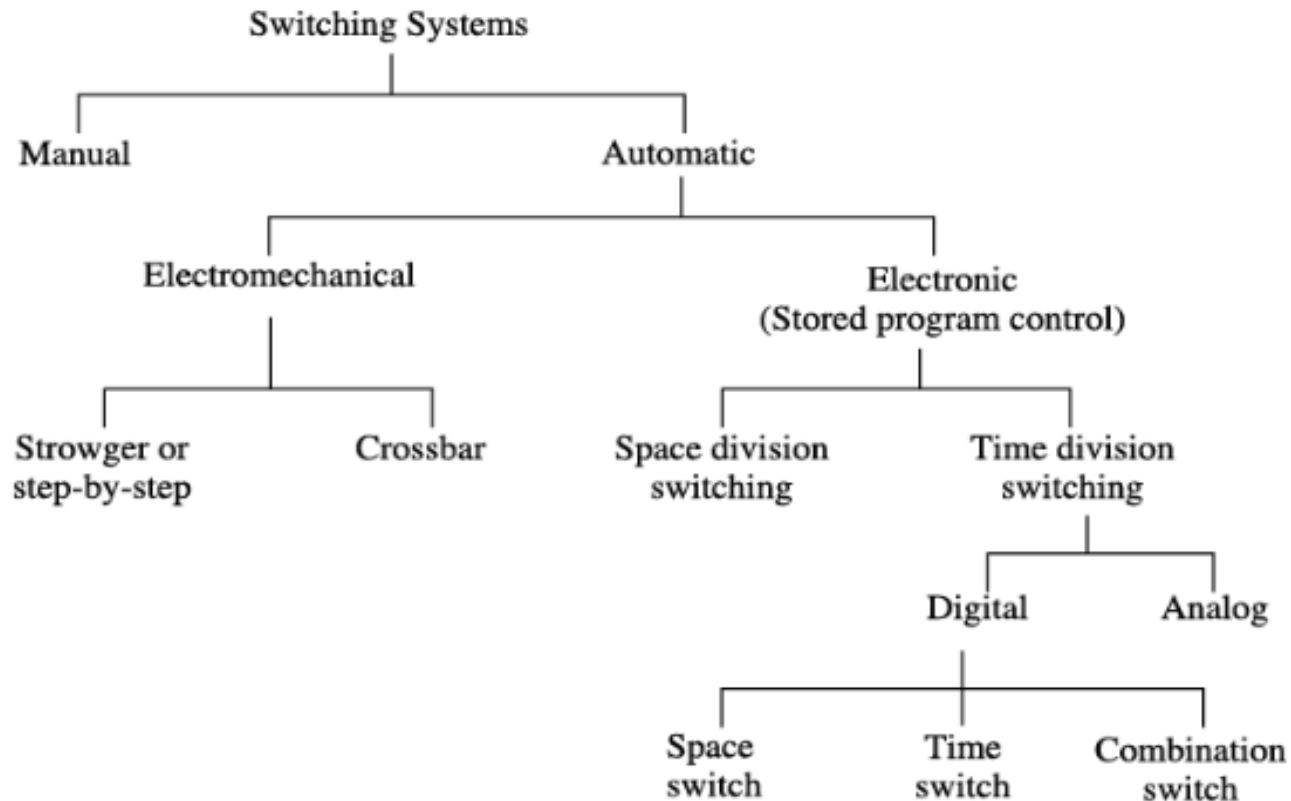


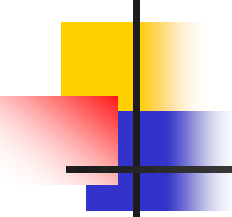
Classification of Switching Systems

- In the early stages of telecommunication systems, the process and stages of switching, played an important to make or break connections.
- At the initial stages, the switching systems were operated manually.
- These systems were later automated.

Classification of Switching Systems

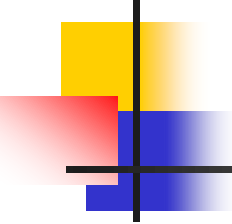
The following flowchart shows the classification of the switching systems

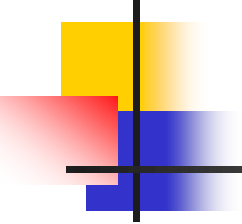




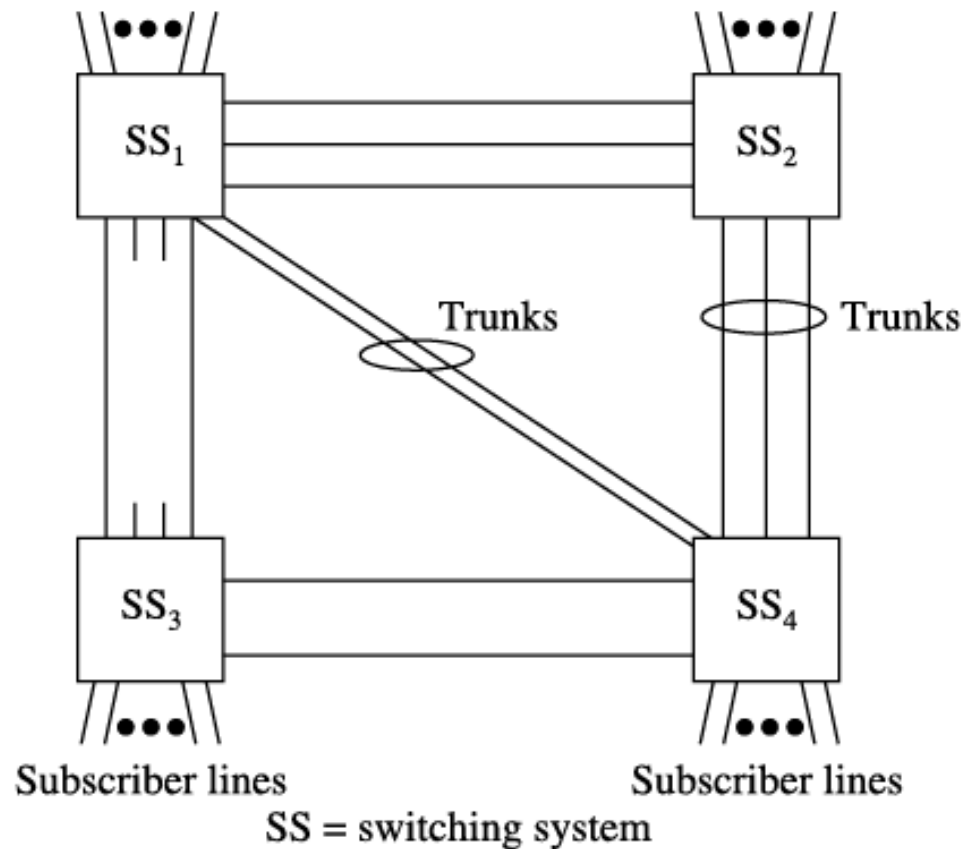
Automatic Switching

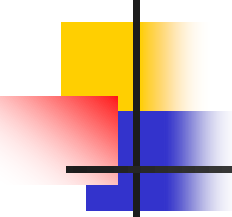
- Automatic switching systems can be classified as **electromechanical and electronic.**
- Electromechanical switching systems include step-by-step and crossbar systems. The step-by step system is better known as **Strowger switching system .**
- The control functions are disassociated from the switching elements and placed in a separate subsystem called **common control subsystem.**

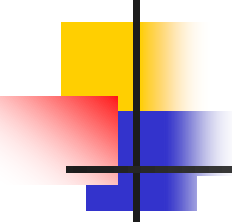
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- In electronic switching systems, the control functions are performed by a computer or a processor. Hence, these systems are called **stored program control** (SPC).
 - The switching scheme used by electronic switching systems may be either **space division switching** or **time division switching**.
 - In analog switching, the sampled voltage levels are transmitted as they are, whereas in digital switching. They are binary coded and transmitted.

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- If the coded values are transferred during the same time interval from input to output, the technique is called **space switching**.
 - If the values are stored and transferred to the output at a later time interval, the technique is called **time switching**.
 - A time division digital switch may also be designed by using a combination of space and time switching techniques.

Switching Network

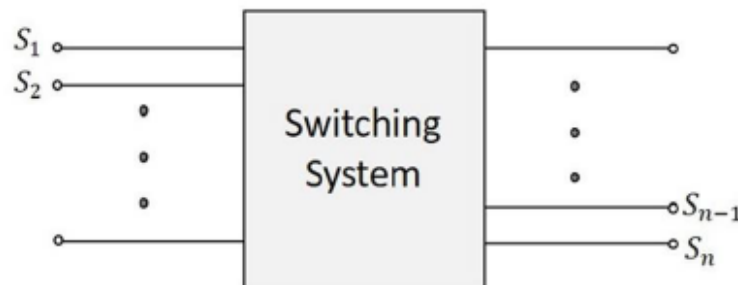


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- The links that run between the switching systems are called **trunks**, and those that run to the subscriber premises are known as **subscriber or local lines**.
 - The first commercial intercity connection was operationalized between New York and Boston in 1884.
 - The first transcontinental trunk connected New York city to San Francisco in 1915
 - The first intercontinental connectivity using short wave radio became operational in 1927 between New York and London.

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- A connection between subscribers attached to switching systems SS2 and SS3 needs to be routed via SS1 or SS4 in a hierarchical structure.
 - Every city has a main or central exchange which is used for routing intercity traffic.
 - Exchanges in different localities of the city are connected to the central exchange either directly or via other exchanges.
 - The central exchange may also be used to route traffic between two exchanges in the same city. Internationally standardised switching hierarchy.

Switching Systems Operation

- Network connection cannot be simply made with telephone sets and bunch of wires, but a good system is required to make or break a connection.
- This system is known as the **Switching System or the Switching Office or the Exchange**.
- The switching system, which is also called the **Telephone Exchange**



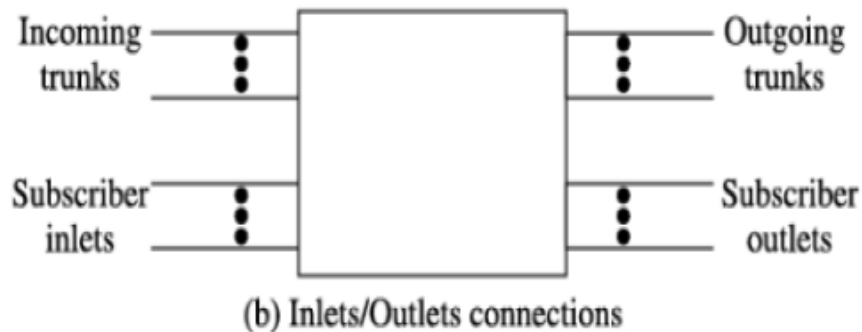
Switching Systems Operation



(a) Model of a switch

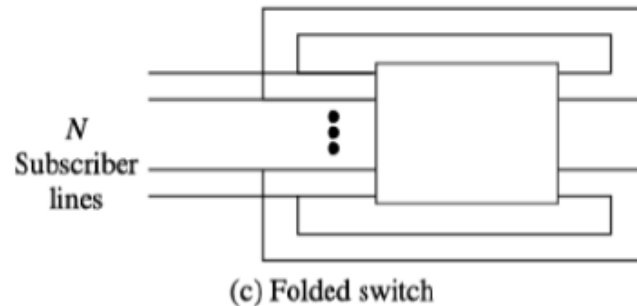
- A major component of a switching system or an exchange is the set of input and output circuits called **inlets** and **outlets**, respectively.
- The primary function of a switching system is to establish an electrical path between a given inlet-outlet pair.
- The hardware used for establishing such a connection is called the **switching matrix** or the **switching fabric**.

Switching Systems Operation



- When $N = M$ the switch is called a symmetric switch.
- The inlets outlets of a switch may be connected to local subscriber lines or to trunks from/to other exchanges as shown in Figure

Switching Systems Operation

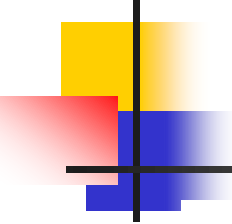


- The output lines are folded back to the input and hence the switch is called a **folded switch**.
- In a switch, all the inlet outlet connections may be used for inter exchange transmission.
- In such a case, the exchange does not support local subscribers and is called a **transit exchange**.



Switching Systems Operation

1. Local call connection between two subscribers in the system
2. Outgoing call connection between a subscriber and an outgoing trunk
3. Incoming call connection between an incoming trunk and a local subscriber
4. Transit call connection between an incoming trunk and an outgoing trunk.



Switching Systems Operation

- A folded switch supports Type 1 connections and a non folded switch supports connections of Type 4.
- In a folded switch with N subscribers, there can be a maximum of only N^2 simultaneous calls or information interchanges.
- The switch may be designed to provide $N/2$ simultaneous switching paths, in which case the switch is said to be **Non blocking**.
- **Blocking Switch**
- **Switching capacity**
- **Blocking probability**
- **Fully connectivity or Fully availability**



Switching Systems Operation

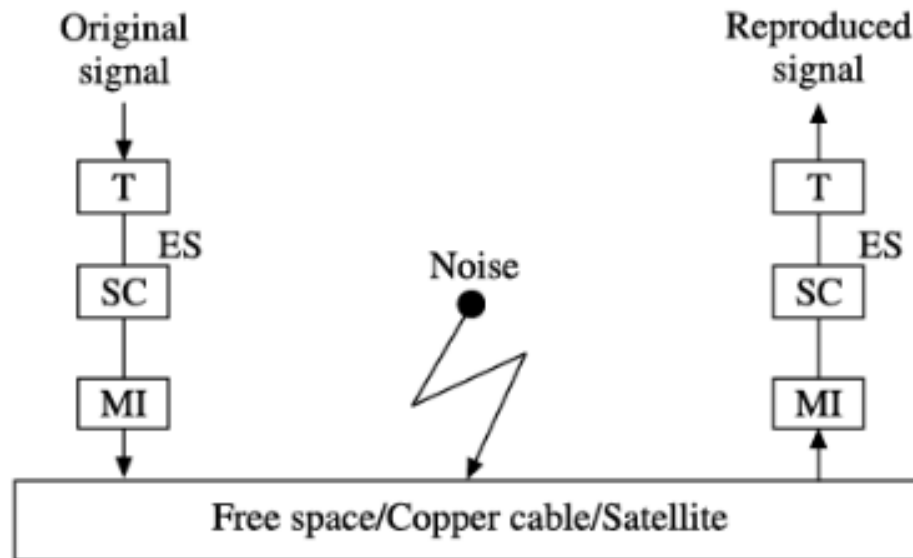
- All the switching exchanges are designed to meet an estimated maximum average simultaneous traffic, usually known as **busy hour traffic**.
- The load on a switching resource is measured in terms of the occupancy of the resource. Such a measure is called the **traffic intensity**
- The measure is obviously dimensionless, but is given a unit known as **erlang (E)**.
- Named after the Danish telephone engineer. A.K. Erlang.



Communication Links

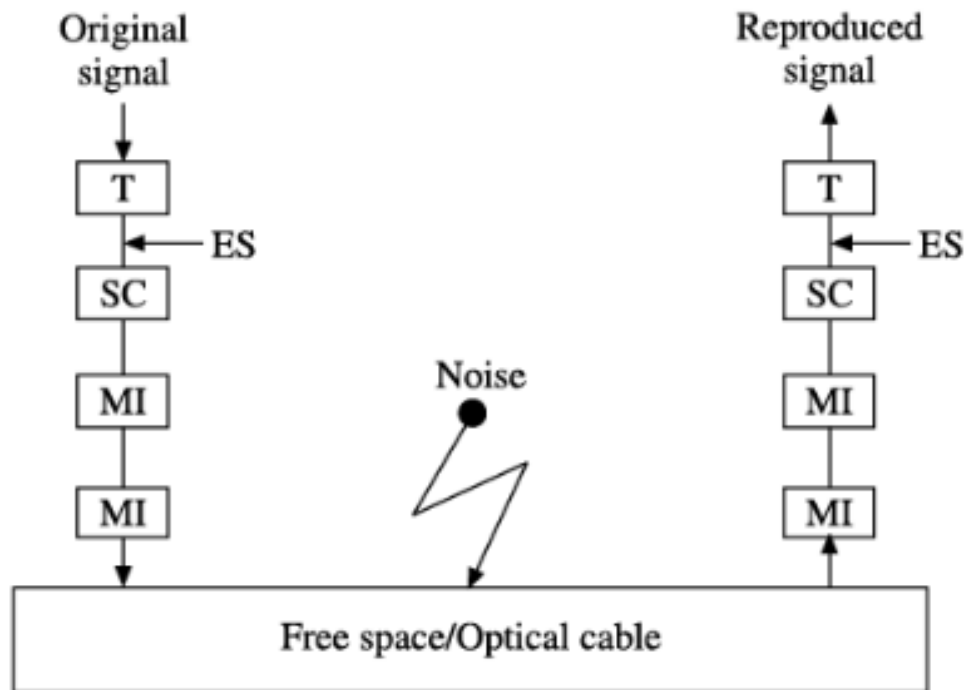
- A telephone switching network is made up of switching systems trunks, subscriber lines and telephone instruments.
- Trunks and subscriber lines are essentially communication links which carry information signals from one point to another.
- There are basically only two forms of communication links electrical and optical.
- In the former information is conveyed by means of electrical energy and in the latter by means of light energy

Communication Links



An electrical communication link

EOC = electrical to optical converter; ES = electrical signal; MI = medium interface
OEC = optical to electrical converter; SC = signal conditioner; T = transducer



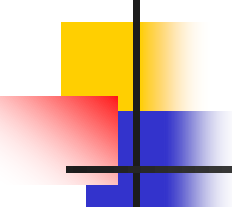
An optical communication link

EOC = electrical to optical converter; ES = electrical signal; MI = medium interface
OEC = optical to electrical converter; SC = signal conditioner; T = transducer



Service Specific Networks

- Telegraph networks
- Telex networks
- Telephone networks
- Data networks



Telegraph networks

- A telegraph is a communications system in which information is transmitted over a wire.
- The basic components include a source of direct current, a length of wire or cable, and a current-indicating device such as a relay, buzzer, or light bulb.
- The term comes from the Greek words "tele," meaning "at a distance" and "graphy," meaning "to write."

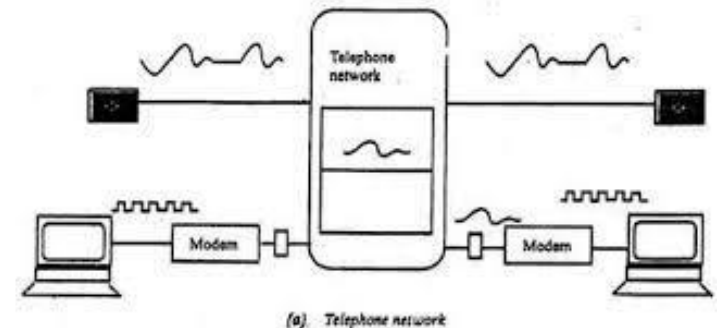
Telex networks

- The **telex** network was a public switched network of teleprinters similar to a telephone network, for the purposes of sending text-based messages. **Telex** was a major method of sending written messages electronically between businesses in the post-World War II period.



Telephone networks

- A **telephone network** is a telecommunications **network** used for **telephone** calls between two or more parties.
- A landline **network** where the **telephones** must be directly wired into a single **telephone** exchange.

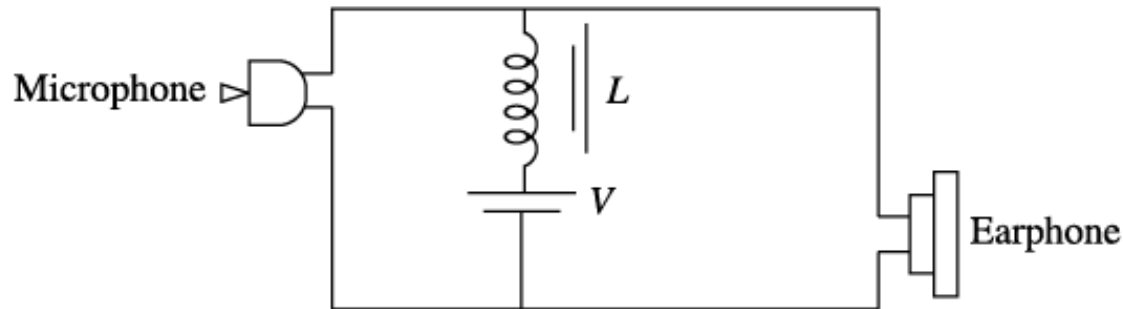


Data networks

- A **data network** is a system that transfers **data** between **network** access points (nodes) through **data** switching, system control and interconnection transmission lines.
- **Data networks** are primarily designed to transfer **data** from one point to one or more points (multipoint).

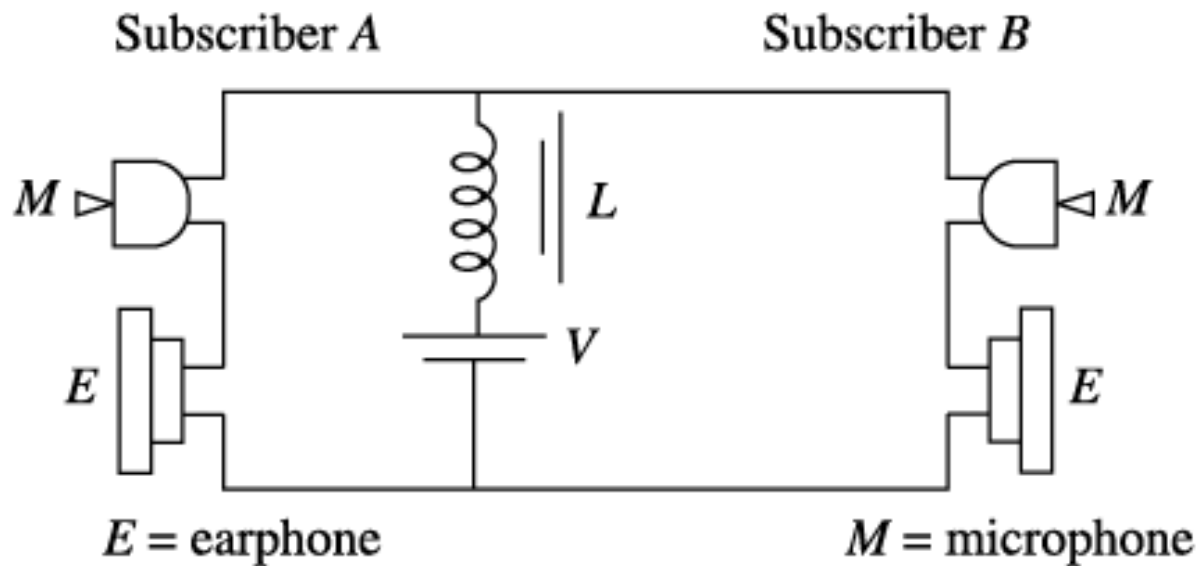


Simple telephone communication



- One way communication.
- Also known as Simplex communication .
- Micro phone and Earphones are transducer of the telecommunication system .
- Micro phone converts speech signal into electrical signal, earphone converts electrical signal into audio signal .

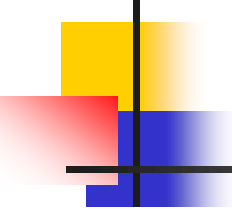
Simple telephone communication





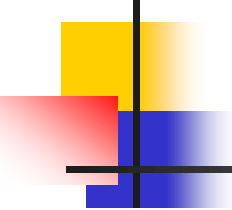
Simple telephone communication

- In a normal telephone communication system information is transferred both ways.
- An entity is capable of both receiving and sending although these do not take place simultaneously.
- Such a form of communication where the information transfer takes place both ways but not simultaneously is known as **half-duplex communication**.
- If the information transfer takes place in both directions simultaneously, then it is called **full-duplex communication**.



Manual Switching

- A microphone requires to be energised in order to produce electrical signals corresponding to the speech waveform.
- In the very early switching systems, the microphone was energised using a battery at the subscriber end. Later, a battery located at the exchange was used.



Manual Switching

- With the advent of automatic switching systems, the manual exchanges have almost gone out of use.
- Today, operator assistance is required on a routine basis, only to connect the incoming calls at a private automatic branch exchange (PABX) to the required extension numbers.

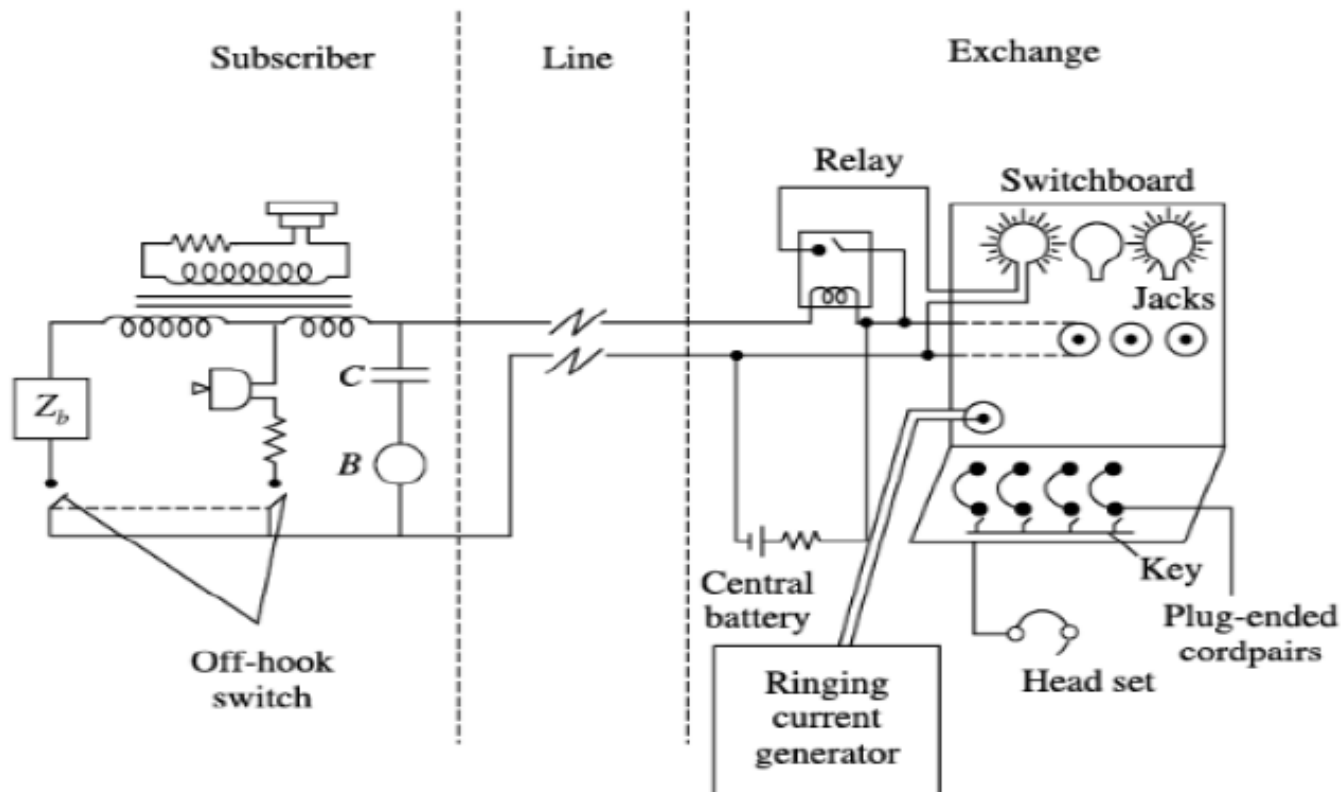


Manual Switching

- Accordingly, one may place the exchanges in one of **two categories**:
 - ✓ Local battery (LB) exchanges
 - ✓ Central battery (CB) exchanges.
- In the LB systems, dry cells were used in subscriber sets to power the microphone.
- LB subscriber sets were provided with a magneto generator.
- In this case, a subscriber needed to rotate a handle to generate the required alternating current to operate indicators at the exchange.
- The necessity to replace dry cells frequently and the cumbersome procedure of rotating the magneto generator led to the development of CB exchanges.

Manual Switching

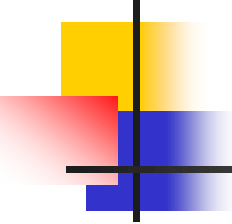
Manned central battery exchange.



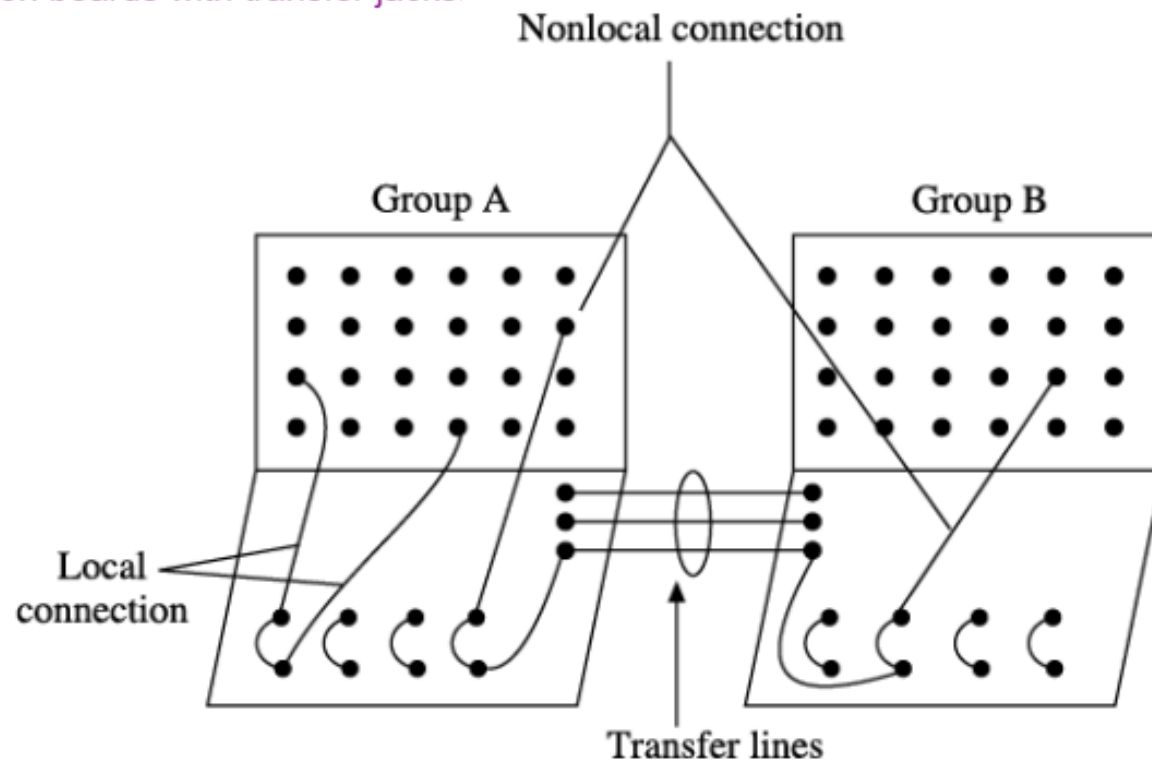


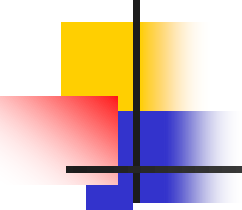
Manual Switching

- The system consists of one or more switchboards manned by operators.
- The subscriber lines are terminated on jacks mounted on the switchboard.
- There is one jack for every subscriber line.
- Associated with each jack is a light indicator to draw the attention of the operator.

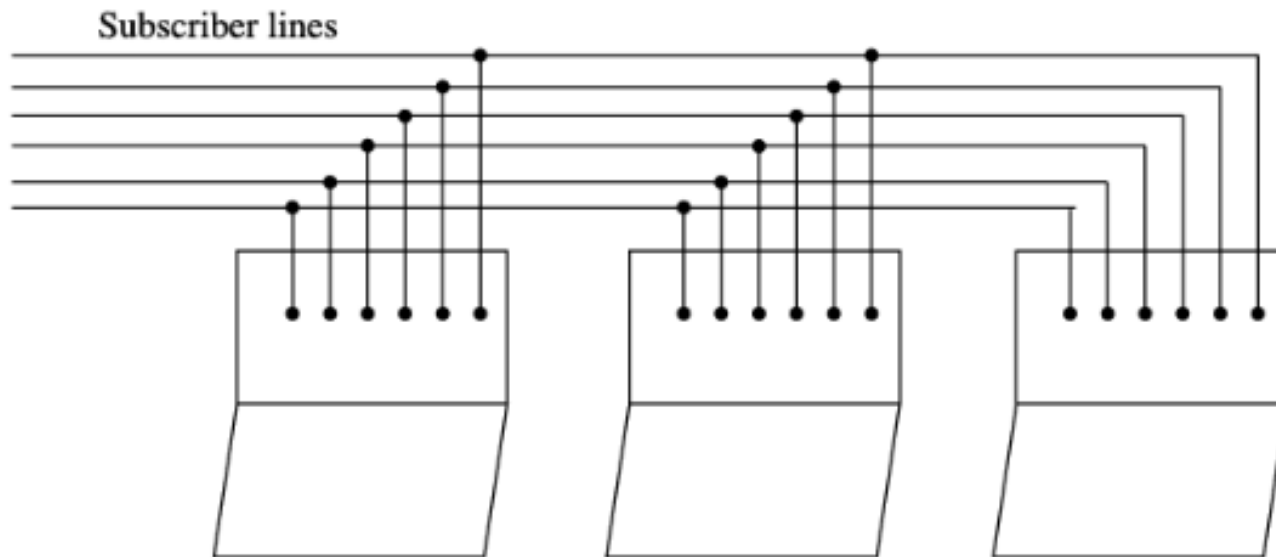
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- When a subscriber lifts the hand set, the off-hook switch is closed, causing a current to flow through the handset and the lamp relay coil.
 - The lamp relay operates and the indicator corresponding to the subscriber lights up.
 - The operator establishes contact with the subscriber by connecting the head set to the subscriber line via the head set key and a plug-ended cord pair.

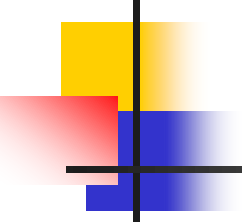
Single termination boards with transfer jacks.

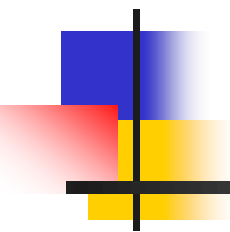




Multitermination boards.



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- When the number of subscribers increases, multiple switchboards and operators are required to handle the traffic.
 - In this case, the subscriber switchboards at the exchange may be of two types:
 - ✓ Single termination switchboards
 - ✓ Multi termination switchboards.



Thank You!